

Project: Signal and Image Processing

Numerical Linear Algebra (1TD452), 2026

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In this project, we study applications of the generalized singular value decomposition (GSVD) and iterative methods to inverse problems in signal and image processing. Many problems, such as deblurring, denoising, and image reconstruction, can be formulated as ill-posed linear inverse problems. In such problems, direct least-squares solutions are unstable because noise in the data is amplified by the ill-conditioning of the forward operator. Regularization is therefore essential. We consider regularized least-squares problems of the form

$$\min_x \|Ax - b\|_2^2 + \lambda^2 \|Lx\|_2^2,$$

where A is the imaging operator and L is a regularization operator. The first part of the project develops the theory of the GSVD, derives the regularized solution, explains its filtering interpretation, and applies it to signal and image deblurring. The second part investigates Krylov subspace methods for solving the same problem on a much larger scale. Finally, the third part considers the variant with an ℓ_1 regularization term,

$$\min_x \|Ax - b\|_2^2 + \lambda \|Lx\|_1,$$

which is better suited for sparse, edge-preserving reconstructions. A fixed-point iterative method for solving this nonsmooth problem is developed and implemented.

1 Part I: Direct methods

Let $x \in \mathbb{R}^n$ denote the unknown true signal or image, $b \in \mathbb{R}^m$ the observed data, and $A \in \mathbb{R}^{m \times n}$ the forward operator describing the imaging system or a suitable discretization of it. A standard observation model is

$$b = Ax + e, \tag{1.1}$$

where e represents measurement noise. In many practical cases, the matrix A is *ill-conditioned* or nearly rank-deficient. Examples include deblurring (or deconvolution problems), tomographic reconstruction, and super-resolution. In such cases, solving the least-squares problem

$$\min_x \|Ax - b\|_2^2 \quad (1.2)$$

is not reliable: tiny perturbations in b may cause large changes in the computed solution. To stabilize the reconstruction, one introduces regularization. A common model is generalized Tikhonov regularization,

$$\min_x \|Ax - b\|_2^2 + \lambda^2 \|Lx\|_2^2. \quad (1.3)$$

The first term is a *fit-to-data term* and the second term is a *regularization term* where L is a matrix encoding prior smoothness or structural information, and $\lambda > 0$ is the regularization parameter. The choice of L determines what features of the image are penalized. For example:

- $L = I$: penalizes large pixel values;
- L first-order difference matrix: penalizes large gradients;
- L second-order difference matrix: penalizes curvature.

In a more generalized case, one can use other norms, for example $\|Ax - b\|_p^p$ for the fitting term and $\|Lx\|_q^q$ for regularization term, where $p > 0$ and $q > 0$. The choice of p is dictated by the distribution of the noise e (e.g., $p = 2$ for Gaussian white noise, $p = 1$ for impulse noise), while the choices of q and of the regularization matrix L are dictated by prior information available on the true solution. We first use $p = q = 2$, but then we also consider the case $p = 2$ and $q = 1$.

SVD

Before introducing the GSVD, it is useful to recall the ordinary SVD. For any matrix $A \in \mathbb{R}^{m \times n}$, there exist orthogonal matrices $U \in \mathbb{R}^{m \times m}$, $V \in \mathbb{R}^{n \times n}$, and a diagonal matrix

$$\Sigma = \begin{bmatrix} \text{diag}(\sigma_1, \dots, \sigma_r) & 0 \\ 0 & 0 \end{bmatrix} \in \mathbb{R}^{m \times n}, \quad \sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0,$$

such that

$$A = U\Sigma V^T.$$

Here $r = \text{rank}(A) \leq n$. The (norm-minimal) least-squares solution of (1.2) can be expressed via the SVD as

$$y_i = \frac{u_i^T b}{\sigma_i}, \quad i = 1, \dots, r, \quad y_i = 0, \quad i = r + 1, \dots, n, \quad x_{\text{LS}} = Vy,$$

or simply

$$x_{\text{LS}} = \sum_{i=1}^r \frac{u_i^T b}{\sigma_i} v_i.$$

This formula shows the main difficulty: when σ_i is very small (the matrix A is ill-conditioned), the coefficient $(u_i^T b)/\sigma_i$ may become very large, which strongly amplifies noise (in b). This is the source of instability in inverse problems. Regularization modifies the coefficients so that terms associated with small singular values are attenuated.

One simple approach is to truncate the small singular values. Assume that we remove singular values smaller than a threshold δ and remain with $k \leq r$ singular values $\sigma_1 \geq \dots \geq \sigma_k > \delta$. Then the matrix $A_k = U_k \Sigma_k V_k^T$ of rank k is the best approximation of A among all $m \times n$ matrices of rank k with minimum norm

$$\|A - A_k\|_2 = \sigma_{k+1} \leq \delta.$$

Here, U_k is a $m \times k$ matrix obtained by keeping the first k columns of U and removing the others. The same for V_k . Σ_k is $k \times k$ principal submatrix of Σ . We then solve the least squares problem with A replaced by A_k , and get the least squares solution

$$x_{\text{TLS}} = \sum_{i=1}^k \frac{u_i^T b}{\sigma_i} v_i. \tag{1.4}$$

which is called the *truncated* least squares solution.

Another approach is to regularize the problem by adding a regularization term and work with (1.3) instead of (1.2).

Task 1. In problem (1.3), assume that $L = I$ and A is full-rank. Prove that the regularized least squares solution can be obtained from normal equation

$$(A^T A + \lambda^2 I) x_{\text{RLS}} = A^T b.$$

In terms of SVD of A , show that the solution reads as

$$x_{\text{RLS}} = Vy, \quad y_i = \frac{\sigma_i}{\sigma_i^2 + \lambda^2} u_i^T b, \quad i = 1, \dots, n.$$

where u_i represents the columns of U . In this case if we define filter factors of the form

$$\phi_i(\lambda) = \frac{\sigma_i^2}{\sigma_i^2 + \lambda^2},$$

then the solution can be written as

$$y_i = \frac{\phi_i(\lambda)}{\sigma_i} u_i^T b.$$

Interpret this filter and this representation of the solution.

Task 2. It is also possible to write the truncated least squares solution (1.4) in terms of filter factors. Define the corresponding filters ϕ_i .

However, when $L \neq I$, the ordinary SVD is no longer sufficient. One needs a decomposition involving both A and L , which leads to the GSVD.

Generalized SVD

Theorem 1. Let $A \in \mathbb{R}^{m \times n}$ and $L \in \mathbb{R}^{p \times n}$. Assume that $m \geq n$. Assume further that

$$\text{rank}(M) = r \leq n, \quad M = \begin{bmatrix} A \\ L \end{bmatrix}.$$

There exist orthogonal matrices $U \in \mathbb{R}^{m \times m}$ and $V \in \mathbb{R}^{p \times p}$, and invertible matrix $W \in \mathbb{R}^{n \times n}$ such that

$$A = U \Sigma_A W^{-1}, \quad L = V \Sigma_L W^{-1}, \quad (1.5)$$

where

$$\Sigma_A = \begin{bmatrix} \text{diag}(\alpha_1, \dots, \alpha_r) & 0 \\ 0 & 0 \end{bmatrix} \in \mathbb{R}^{m \times n}, \quad \Sigma_L = \begin{bmatrix} \text{diag}(\beta_1, \dots, \beta_r) & 0 \\ 0 & 0 \end{bmatrix} \in \mathbb{R}^{p \times n}$$

with $0 \leq \alpha_1 \leq \dots \leq \alpha_{k+1} = \dots = \alpha_r = 1$, and $1 \geq \beta_1 \geq \dots \geq \beta_{k+1} = \dots = \beta_r = 0$ where $k = \min\{p, r\}$. (The last $r - k$ singular values α_j are 1, and the last $r - k$

singular values β_j are 0). We also have

$$\alpha_i^2 + \beta_i^2 = 1, \quad i = 1, 2, \dots, r$$

The ratios

$$\gamma_i = \frac{\alpha_i}{\beta_i}, \quad i = 1, 2, \dots, r$$

are called the generalized singular values of pair (A, L) , and the decomposition (1.5) is called GSVD of pair (A, L) .

A proof for Theorem 1 uses another decomposition called *CS decomposition*. For more details see Chapter 6 of [2, 2013]. If $L = I$, then the GSVD reduces to the ordinary SVD of A by setting $W = V$. In that sense the GSVD is the generalization of SVD to the pair (A, L) .

Generalized Tikhonov Regularization

Task 3. Consider the optimization problem (1.3). Show that the solution of this problem satisfies normal equations

$$(A^T A + \lambda^2 L^T L)x = A^T b. \quad (1.6)$$

Under what condition the solution is unique?

In principle one can solve this linear system directly. However, the GSVD gives a more stable algorithm and more informative representation that explains the action of regularization mode by mode.

Task 4. Using GSVD of matrix pair (A, L) show that the solution to regularized least squares problem (1.3) is given by

$$x = Wy, \quad y_i = \frac{\alpha_i}{\alpha_i^2 + \lambda^2 \beta_i^2} (u_i^T b). \quad (1.7)$$

Write the formula (1.7) in terms of *filter* factors $\phi_i(\lambda) \in [0, 1]$, and interpret the solution. How the filter regularized the solution by removing the unstable components?

Signal and image processing

In signal and image processing, the matrix A models the acquisition system, while L encodes prior assumptions about the signal or image.

Deblurring: Deblurring can be formulated as an integral equation of the form

$$\int_{\mathbb{R}^d} K(s, t) x(t) dt + e(s) = b(s) \quad (1.8)$$

where $t, s \in \mathbb{R}^d$ represent spatial information. The kernel $K(s, t)$, also known as the point spread function (PSF), defines the blur. The kernel is often shift-invariant, i.e. $K(s, t) \equiv K(s - t)$, which makes the equation (1.8) a deconvolution problem. In practical settings, discrete data are collected in finite regions, so that the continuous model (1.8) is discretized and yields a linear system of equation of the form (1.1). In this particular case, the matrix A represents the blurring operator, which is defined starting from the PSF and the boundary conditions on the unknown quantity of interest. As an example PSF can be given by a rescaled, possibly asymmetric Gaussian of the form

$$K(s, t) = c \exp \left(- (s - t)^T B (s - t) \right), \quad c > 0 \quad (1.9)$$

and B is a positive definite covariance matrix, mostly a diagonal matrix of the form $B = \text{diag}(b_1, \dots, b_d)$.

Choice of the regularization matrix: The choice of L is problem dependent: The simplest regularization is obtained for $L = I_n$, the identity matrix of size $n \times n$, which suppresses large amplitudes uniformly.

Another choice for L is a finite-difference discretization of the first derivative operator. Let

$$L = \begin{bmatrix} 1 & -1 & & & \\ & 1 & -1 & & \\ & & \ddots & \ddots & \\ & & & 1 & -1 \end{bmatrix} \in \mathbb{R}^{(n-1) \times n}$$

be a rescaled finite-difference discretization of the first derivative operator in a 1D case. For problems that depend on two spatial dimensions (x, y) the matrix L is used to obtain discretizations of the first derivatives in the each direction x and y . We name them L_1 and L_2 respectively. For a static image represented as a 2D array $X \in \mathbb{R}^{n_1 \times n_2}$, such that

$$x = \text{vec}(X) \in \mathbb{R}^n, \quad n = n_1 n_2,$$

is obtained by stacking the columns of X , its horizontal and vertical derivatives are given as

$$\begin{aligned}\text{vec}(L_1 X) &= (I_{n_2} \otimes L_1)x \in \mathbb{R}^{n_2(n_1-1)} \\ \text{vec}(X L_2^T) &= (L_2 \otimes I_{n_1})x \in \mathbb{R}^{(n_2-1)n_1},\end{aligned}$$

respectively, where $\otimes$ is the Kronecker product, and I_n is the identity matrix of size $n \times n$. The discrete gradient is then expressed as

$$L_G = \begin{bmatrix} I_{n_2} \otimes L_1 \\ L_2 \otimes I_{n_1} \end{bmatrix}.$$

Task 5. In this task, we consider a one-dimensional deblurring problem arising in signal processing. Let $x(t)$ denote the true signal on a finite interval $[a, b]$. Sample the signal at n equidistant points $t_1, \dots, t_n \in [a, b]$, and denote the resulting discrete signal by the vector x .

Construct the matrix A by discretizing the integral operator (1.8) on $[a, b]$ using the box quadrature rule at the sample points t_i . Use the Gaussian kernel with $B = \beta$, where $\beta > 0$ is chosen sufficiently large.

Assume that the observed data are given by $b = Ax + e$, where $e = \delta \mathcal{N}(0, 1)$ is a Gaussian noise vector with noise level δ .

Reconstruct the signal by solving the regularized least-squares problem (1.3) using the GSVD of the pair (A, L) , where L is a finite-difference approximation of the first derivative.

Carry out the following experiments.

1. Compute both the regularized solution and the unregularized least-squares solution. Plot the true signal, the blurred noisy data, and the reconstructed signals in the same figure. Compare the quality of the reconstructions.
2. Plot the generalized filter factors $\phi_i(\lambda)$ and discuss how they suppress the components associated with small generalized singular values. Explain how the filter changes as λ varies.
3. Repeat the experiment for different values of the regularization parameter λ . Investigate the effect of under-regularization and over-regularization, and determine a value of λ that gives a good compromise between noise suppression and reconstruction accuracy.

4. Repeat the experiment for several noise levels. For each noise level, compare the regularized and unregularized solutions, and discuss how the quality of the reconstruction depends on the amount of noise. Explain why regularization becomes increasingly important as the noise level increases.
5. Search the literature for techniques used to select an appropriate regularization parameter λ , such as the discrepancy principle, the L-curve criterion, generalized cross-validation (GCV), or the quasi-optimality criterion. Implement one of these methods and compare the automatically selected value of λ with your manually chosen value.

The Python implementation of the GSVD is available on the course webpage. Download and use it in your implementation.
For all experiments use

$$x(t) = \exp(t) \sin(\pi t), \quad t \in [-1, 1].$$

2 Part II: Iterative methods

The GSVD provides a powerful theoretical framework for understanding generalized Tikhonov regularization and leads to efficient algorithms for small and medium-sized problems. However, computing a GSVD requires dense matrix factorizations whose computational cost is $\mathcal{O}(n^3)$ and whose memory requirements are $\mathcal{O}(n^2)$. Consequently, the GSVD quickly becomes impractical for large-scale applications such as image restoration, tomographic reconstruction, and other inverse problems arising in scientific computing.

In this part of the project we consider iterative Krylov subspace methods. Unlike direct methods, these algorithms require only matrix-vector products with the matrices A and A^T , making them suitable for very large sparse problems. Throughout this part we assume that the blurring matrix A and the regularization matrix L are stored as *sparse* matrices.

Consider a grayscale image represented by a matrix

$$X_{\text{true}} \in \mathbb{R}^{n_1 \times n_2},$$

and let

$$x_{\text{true}} = \text{vec}(X_{\text{true}}) \in \mathbb{R}^n, \quad n = n_1 n_2,$$

denote the vectorized image. The blurred and noisy image is modeled by

$$b = Ax_{\text{true}} + e,$$

where A represents the blurring operator and e is Gaussian white noise. Use the Gaussian point spread function (1.9) where B is positive definite. For an isotropic blur, one may take

$$B = \beta I.$$

For regularization, use the discrete gradient

$$L_G = \begin{bmatrix} I_{n_2} \otimes L_1 \\ L_2 \otimes I_{n_1} \end{bmatrix},$$

where L_1 and L_2 are first-order finite-difference matrices in the two spatial directions. Use this experiment in all subsequent tasks unless otherwise stated.

Task 6. Download the image from the course page. Resize the image to approximately 300×300 pixels and convert it to grayscale.

Construct the blurring matrix A corresponding to the Gaussian kernel (1.9) in the 2D case. Let $\varepsilon > 0$ be a prescribed threshold and replace every matrix entry satisfying

$$|a_{ij}| < \varepsilon$$

by zero. Generate blurred noisy data according to

$$b = Ax + e,$$

where $e = \sigma \mathcal{N}(0, 1)$ is Gaussian white noise where σ is the noise level.

Also construct the regularization matrix $L = L_G$. Store the resulting matrices using a sparse matrix format. For some values of β and ε :

1. Plot the sparsity pattern of A and compute the percentage of nonzero entries.
2. Do the same for L .
3. Estimate the memory required by the dense and sparse representations.
4. Discuss how the parameters influence sparsity and the computational cost.

Task 7. Consider again the regularized least-squares problem

$$\min_x \|Ax - b\|_2^2 + \lambda^2 \|Lx\|_2^2.$$

In Task 3, you showed that the solution satisfies the normal equations

$$(A^T A + \lambda^2 L^T L)x = A^T b. \quad (2.1)$$

Assume that

$$\text{rank} \begin{bmatrix} A \\ L \end{bmatrix} = n.$$

Prove that the matrix

$$A^T A + \lambda^2 L^T L$$

is symmetric positive definite. Explain why this property makes the conjugate gradient (CG) method applicable.

The proof should be written for general matrices A and L , not just for the above image matrices.

Task 8. Implement the CG algorithm for solving the regularized normal equations (2.1). The algorithm should benefit from the sparse matrix representations of both A and L without forming any matrix-matrix products.

For the above blurring matrix A and regularization matrix $L = L_G$ investigate the influence of

1. the regularization parameter λ ,
2. the stopping tolerance,
3. the maximum number of iterations.

For each experiment, plot

- the original and blurred images, and the reconstructed image,
- the relative residuals of the CG algorithm,
- the relative reconstruction error,
- the objective function

$$\|Ax - b\|_2^2 + \lambda^2 \|Lx\|_2^2$$

as functions of the iteration number.

Discuss the convergence behavior.

Task 9. Solve the same regularized problem using MINRES and compare CG and MINRES with respect to

- convergence rate,
- computational time,
- number of matrix-vector multiplications.

You need to implement the MINRES algorithm special for the regularized problem (1.6). Explain why CG is generally expected to outperform MINRES for this particular problem.

There is one subtlety. Although we do not form the normal equation matrix, CG and MINRES are still mathematically solving the normal equations. Therefore, the conditioning issue remains:

$$\kappa(A^T A) = \kappa(A)^2$$

One approach is to use a proper preconditioner for the system.

Task 10. Construct a suitable preconditioner for your CG method (for example, an incomplete Cholesky factorization or another appropriate sparse preconditioner). Derive the corresponding preconditioned CG algorithm and use it to solve the regularized least-squares problem. Compare the performance of CG and PCG with respect to

- the number of iterations required to reach convergence,
- the computational time,
- the relative residual,
- the quality of the reconstructed solution.

An alternative and sometimes more efficient algorithm is the LSQR method which is based on Golub–Kahan/Lanczos bidiagonalization.

Task 11. Read about the LSQR algorithm. Explain why LSQR can be applied on regularized problem (1.6) and how it avoids explicitly forming the matrix $A^T A$ or $L^T L$. Compare LSQR with CG. Discuss the numerical advantages and disadvantages of the two methods for large ill-conditioned least-squares problems.

Task 12. One remarkable feature of iterative methods for ill-posed inverse problems is that the iteration number itself acts as a regularization parameter. For both CG and LSQR,

1. plot the reconstruction error versus the iteration number,
2. determine the iteration giving the smallest reconstruction error,
3. investigate how this optimal iteration depends on the noise level.

Explain why continuing the iterations beyond the optimal point often deteriorates the reconstruction quality.

Task 13. Summarize the differences between direct methods (SVD/GSVD) and iterative Krylov methods for solving inverse problems. Your discussion should include

- computational complexity,
- memory requirements,
- numerical stability,
- scalability,
- suitability for small and large problems,
- advantages and disadvantages of each approach.

3 Part III: Extension to ℓ_1 regularization

The quadratic penalty $\|Lx\|_2^2$ promotes global smoothness, but it tends to oversmooth edges. In image processing, edges are important features, so one often replaces the ℓ_2 penalty by an ℓ_1 penalty:

$$\min_x \frac{1}{2} \|Ax - b\|_2^2 + \lambda \|Lx\|_1. \quad (3.1)$$

Here

$$\|Lx\|_1 = \sum_i |(Lx)_i|.$$

This model encourages sparsity of Lx . If L is a discrete gradient operator, then sparsity of Lx means that the image has relatively few large jumps, which is consistent with piecewise smooth images separated by edges.

The problem is still convex, but it is no longer differentiable because the absolute value function is not differentiable at zero. Therefore the linear GSVD formula, and the normal equations derived for the quadratic case no longer apply directly.

Nevertheless, one can still design iterative methods that repeatedly solve easier subproblems. One useful approach is based on fixed-point iteration or *iteratively-reweighted least-squares (IRLS)* approximation.

Fixed-Point iteration for the ℓ_1 -regularized problem

Consider the minimization problem (3.1). A standard idea is to approximate the ℓ_1 -norm by a weighted quadratic term. Since

$$|z| \approx \frac{z^2}{2\sqrt{z^2 + \varepsilon^2}} + \text{constant},$$

for small $\varepsilon > 0$, one can write

$$\|Lx\|_1 \approx \frac{1}{2} \sum_i w_i(x) (Lx)_i^2 + \text{constant},$$

where the weights are defined by

$$w_i(x) = \frac{1}{\sqrt{(Lx)_i^2 + \varepsilon^2}}.$$

This leads to an iterative scheme. Given the current iterate $x^{(k)}$, define the diagonal weight matrix

$$W^{(k)} = \text{diag} \left(\frac{1}{\sqrt{(Lx^{(k)})_i^2 + \varepsilon^2}} \right). \quad (3.2)$$

Then compute the next iterate $x^{(k+1)}$ as the solution of the quadratic problem

$$\min_x \frac{1}{2} \|Ax - b\|_2^2 + \frac{\lambda}{2} (Lx)^T W^{(k)} (Lx).$$

The corresponding normal equation is

$$(A^T A + \lambda L^T W^{(k)} L) x^{(k+1)} = A^T b.$$

This defines a fixed-point map

$$x^{(k+1)} = G(x^{(k)}),$$

where $G(x)$ is the solution of

$$(A^T A + \lambda L^T W(x) L) G(x) = A^T b.$$

Thus the method is called a fixed-point iteration: one updates the weights from the current solution estimate and then solves a weighted Tikhonov problem.

From weight matrix (3.2) we observe that where $(Lx^{(k)})_i$ is large, the weight is smaller, so sharp transitions are penalized less; and where $(Lx^{(k)})_i$ is small, the weight is larger, so small oscillations are penalized more. This is exactly why ℓ_1 -type regularization better preserves edges than ℓ_2 -type regularization.

Task 14. Develop a fixed-point iterative algorithm (or write a pseudocode) for solving the ℓ_1 -regularized least-squares problem (3.1). The algorithm should start from an initial guess $x^{(0)}$ and generate a sequence of iterates $x^{(k)}$ by repeatedly solving the corresponding normal equations arising in each iteration. Specify a suitable stopping criterion and terminate the iterations once the prescribed convergence tolerance is satisfied.

Connection to GSVD

At each iteration, the method solves a weighted quadratic problem of the form

$$\min_x \|Ax - b\|_2^2 + \lambda \|\sqrt{W^{(k)}} Lx\|_2^2.$$

Task 15. Show how the GSVD can be applied locally to solve the ℓ_1 -regularized least-squares problem. Derive the resulting algorithm and implement it as a code. Then apply it on the problem of Task 5 and compare the results with those of the ℓ_2 regularization. Analyze the effect of the parameter ε .

Task 16.

When the matrices involved in the L_1 regularization problem are large and sparse, computing a GSVD at every iteration is no longer practical. Investigate how Krylov subspace methods can be used instead.

Implement one of the Krylov subspace methods and compare the results with the ℓ_2 regularization problem on a large-scale image deblurring problem.

Finally, we note that a variety of efficient algorithms have been developed for solving ℓ_1 -regularized optimization problems. These include proximal gradient methods, such as the Iterative Shrinkage-Thresholding Algorithm (ISTA) and its accelerated variant FISTA, the Alternating Direction Method of Multipliers (ADMM), and the split Bregman method. These approaches are beyond the scope of this project, and we do not consider them here. For a comprehensive treatment of these methods, see [1, 3].

References

- [1] Amir Beck. *First-Order Methods in Optimization*. SIAM, 2017.
- [2] Gene H. Golub and Charles F. Van Loan. *Matrix Computations*. Johns Hopkins University Press, 4th edition, 2013.
- [3] Per Christian Hansen. *Discrete Inverse Problems: Insight and Algorithms*. SIAM, Philadelphia, PA, 2010.